ScientistTwo

Pioneering the Human Knowledge Frontier with Autonomous AI

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A problem in. A research cycle out.

Given a scientific problem, ScientistTwo identifies baseline limitations, proposes new ideas, runs experiments, and produces a manuscript with executable code.

Research that learns from feedback

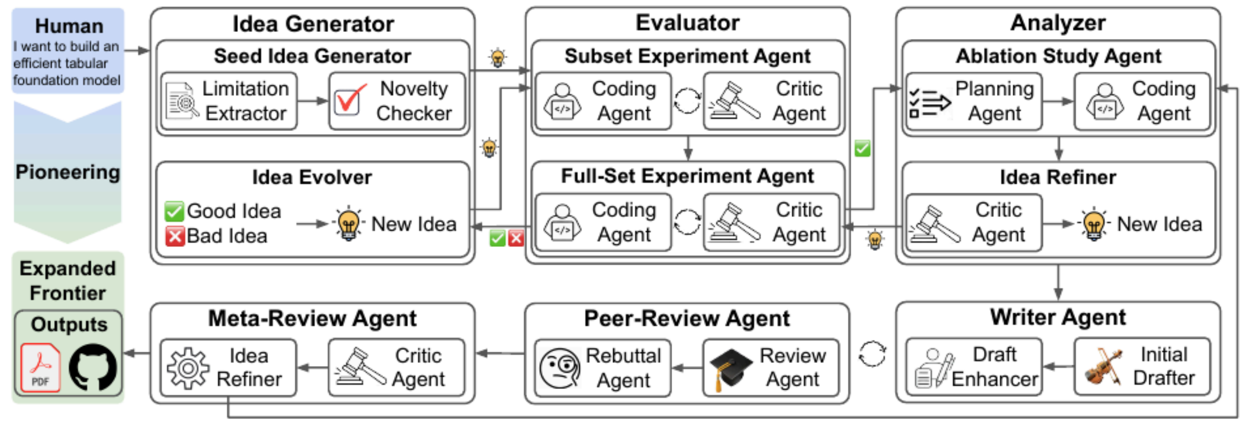


Figure 3 · ScientistTwo connects idea generation, subset-to-full evaluation, ablation analysis, writing, peer review and meta-review in a closed loop.

What drives the loop

Stage	Decision and feedback
Evaluation	Screen on a subset; scale promising ideas to the full benchmark.
Ablation	Isolate component contributions; refine the method and re-test.
Rebuttal	Run supplementary experiments to address reviewer critiques.

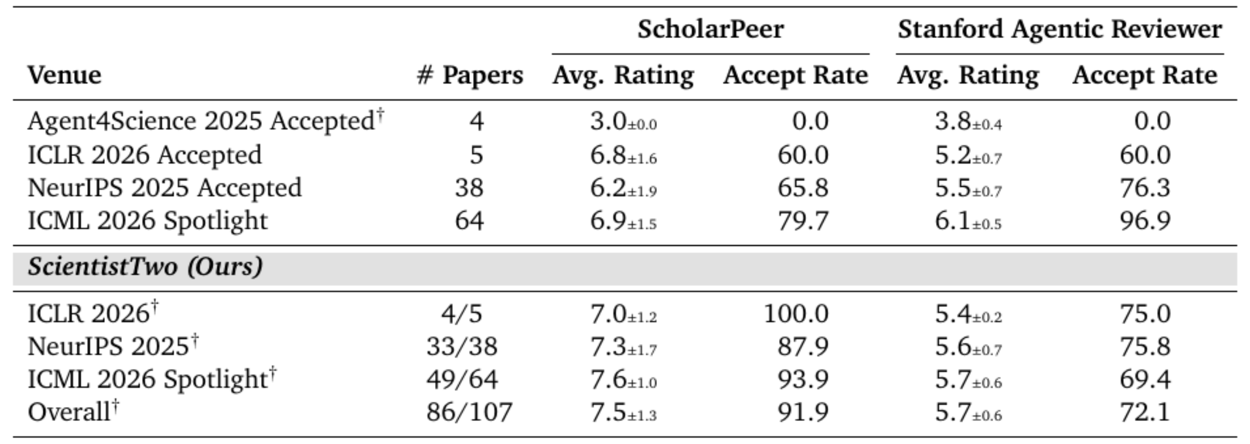
Measured progress, bounded claims

80.4%

ScientistTwo improved 86 of 107 target problems.

25.2%

Mean relative gain over human baselines across successful cases.



Venue	# Papers	ScholarPeer Avg. Rating	ScholarPeer Accept Rate	Stanford Agentic Reviewer Avg. Rating	Stanford Agentic Reviewer Accept Rate
Agent4Science 2025 Accepted†	4	$3.0_{\pm 0.0}$	0.0	$3.8_{\pm 0.4}$	0.0
ICLR 2026 Accepted	5	$6.8_{\pm 1.6}$	60.0	$5.2_{\pm 0.7}$	60.0
NeurIPS 2025 Accepted	38	$6.2_{\pm 1.9}$	65.8	$5.5_{\pm 0.7}$	76.3
ICML 2026 Spotlight	64	$6.9_{\pm 1.5}$	79.7	$6.1_{\pm 0.5}$	96.9
ScientistTwo (Ours)					
ICLR 2026†	4/5	$7.0_{\pm 1.2}$	100.0	$5.4_{\pm 0.2}$	75.0
NeurIPS 2025†	33/38	$7.3_{\pm 1.7}$	87.9	$5.6_{\pm 0.7}$	75.8
ICML 2026 Spotlight†	49/64	$7.6_{\pm 1.0}$	93.9	$5.7_{\pm 0.6}$	69.4
Overall†	86/107	$7.5_{\pm 1.3}$	91.9	$5.7_{\pm 0.6}$	72.1

Table 3 · AI reviewers score human reference papers and ScientistTwo outputs. Ratings use a 1–10 scale; acceptance columns report simulated acceptance rates (%). † marks AI-generated papers.

Two AI reviewers, different roles

Among the 86 successfully generated papers, ScientistTwo reaches 91.9% simulated acceptance under ScholarPeer and 72.1 under Stanford Agentic Reviewer. ScholarPeer guides refinement; Stanford is held out during development. These are AI-review outcomes, not conference decisions.

> Table 4 reports a median gain of 7.7 and mean gain of 25.2 (both in %) for successful ScientistTwo cases. The mean should not be read as a typical gain for every task.

Why iteration matters

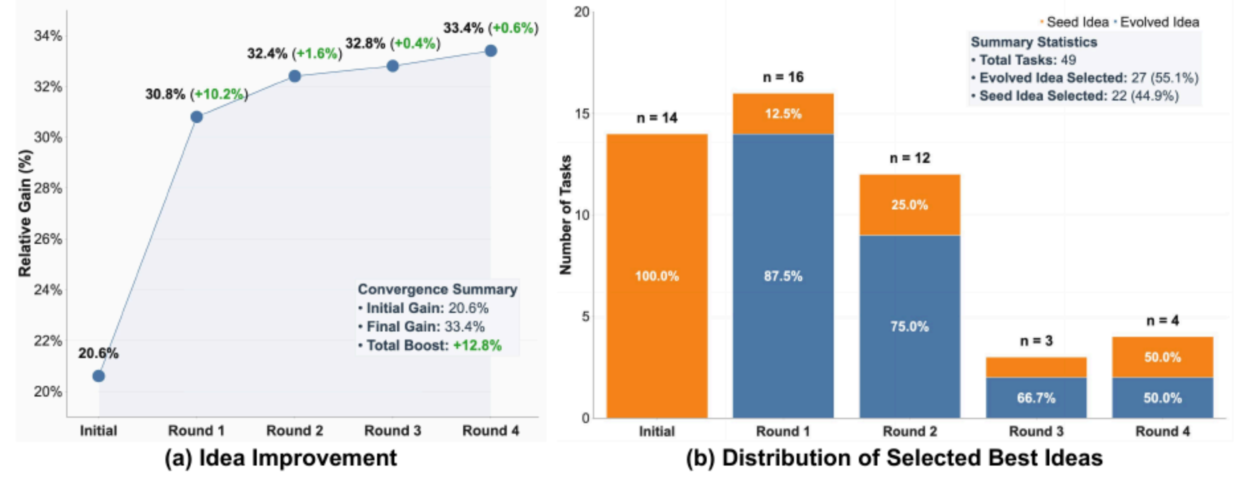


Figure 9 · On ICML-derived tasks, ScientistTwo gains most during early refinement. Evolved ideas dominate later selections, using both successful and failed execution traces as feedback.

A separate human check

Nine experienced reviewers assessed 33 NeurIPS-derived papers. ScientistTwo reached overall parity with human papers: 3.0 on a relative-preference scale where 3.0 means parity. Methodological rigor remained slightly behind.

A promising loop—with limits

ScientistTwo demonstrates autonomous empirical research, but does not yet consistently deliver spotlight-level conceptual breakthroughs. On the 33 NeurIPS-derived tasks used for cost analysis, the reported mean run takes about 2.5 days and costs $3,765, including tokens and virtual machines.

Source: Nam et al. · ScientistTwo · arXiv:2609.19644v1 · Original figures and table retained; explanatory text is editable.